

Chapter 1

Appendix

1.1 Full Transmitter Circuit Schematic

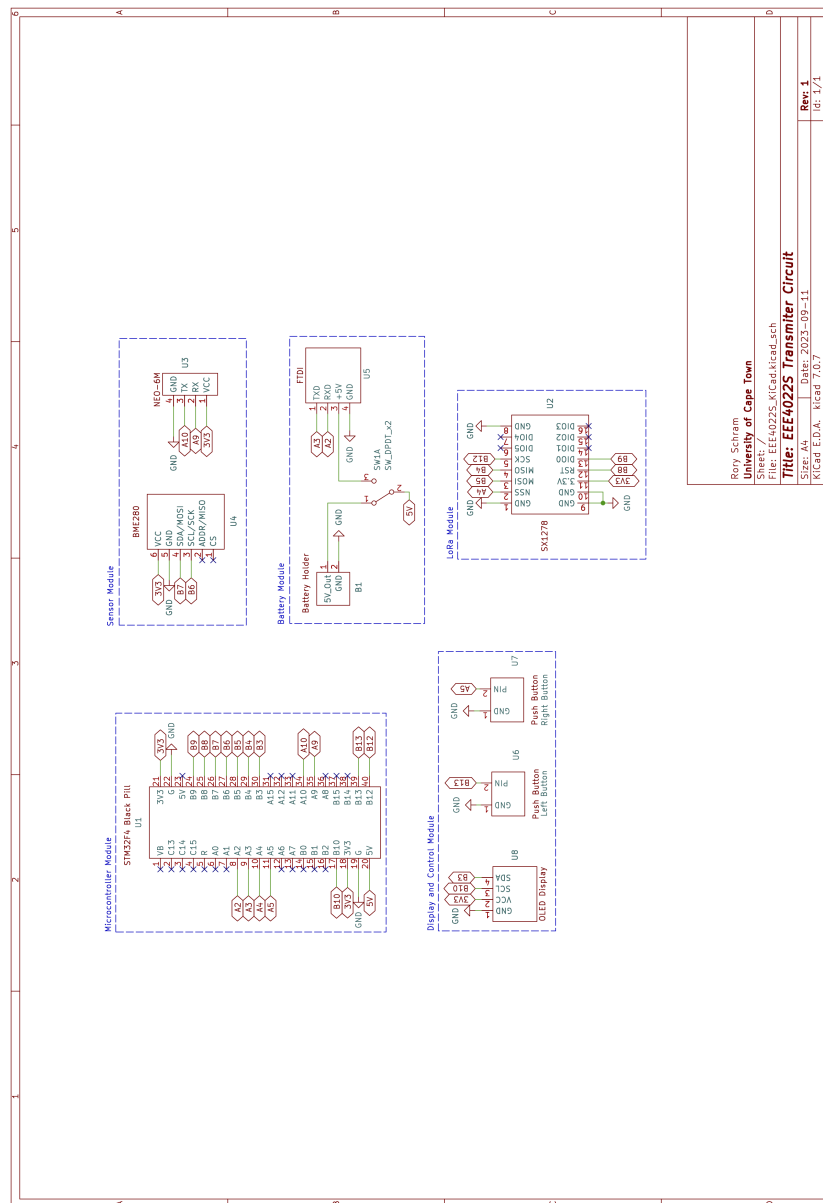


Figure 1.1: Full Transmitter Circuit Schematic

1.2 Full Receiver Circuit Schematic

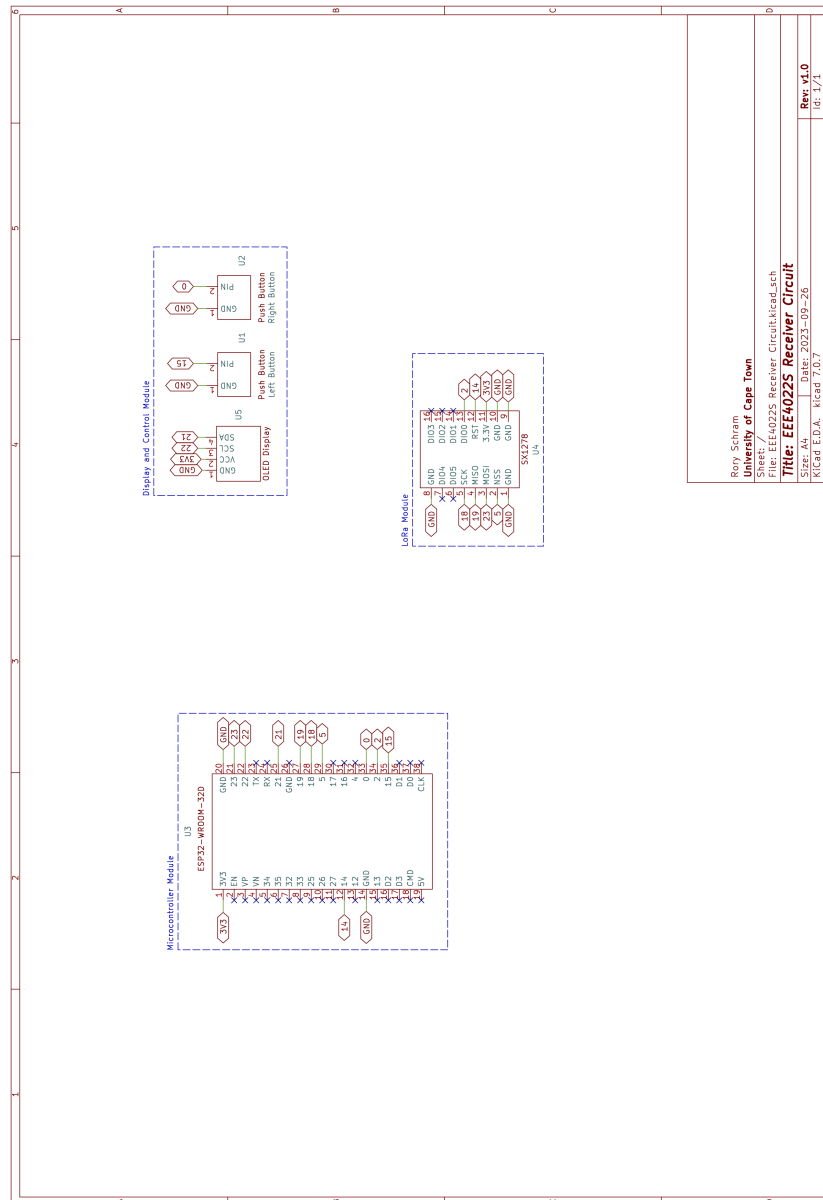


Figure 1.2: Full Receiver Circuit Schematic

1.3 Long Range Test Interference

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Mode: 50
0x26: 0x0
Received packet 'f@VjUUUUjZZZ@YefiZZZZZZZZZZZZ;eZZVZ@UU[' with RSSI -98 and frequency error of -12043
Received packet 'f@VjUUUUjZZZ@YefiZZZZZZZZZZZZ;eZZVZ@UU[' with RSSI -97 and frequency error of -12022
Received packet 'f@VjUUUUjZZZ@YefiZZZZZZZZZZZZ;eZZVZ@UU[' with RSSI -97 and frequency error of -11997
Received packet 'f@VjUUUUjZZZ@YefiZZZZZZZZZZZZ;eZZVZ@UU[' with RSSI -98 and frequency error of -11989
Received packet 'f@VjUUUUjZZZ@YefiZZZZZZZZZZZZ;eZZVZ@UU[' with RSSI -96 and frequency error of -11972
Received packet 'f@VjUUUUjZZZ@YefiZZZZZZZZZZZZ;eZZVZ@UU[' with RSSI -97 and frequency error of -11964
Received packet 'f@VjUUUUjZZZ@YefiZZZZZZZZZZZZ;eZZVZ@UU[' with RSSI -96 and frequency error of -11951
Received packet 'f@VjUUUUjZZZ@YefiZZZZZZZZZZZZ;eZZVZ@UU[' with RSSI -96 and frequency error of -11939
Received packet 'f@VjUUUUjZZZ@YefiZZZZZZZZZZZZ;eZZVZ@UU[' with RSSI -95 and frequency error of -11951
Received packet 'f@VjUUUUjZZZ@YefiZZZZZZZZZZZZ;eZZVZ@UU[' with RSSI -97 and frequency error of -11955
Received packet 'f@VjUUUUjZZZ@YefiZZZZZZZZZZZZ;eZZVZ@UU[' with RSSI -96 and frequency error of -11951
Received packet 'f@VjUUUUjZZZ@YefiZZZZZZZZZZZZ;eZZVZ@UU[' with RSSI -97 and frequency error of -11968
Average RSSI: -96.67

Mode: 51
0x26: 0x8

Mode: 52
0x26: 0x8

Mode: 53
0x26: 0x8
Received packet 'f@VjUUUUjZZZ@YefiñagøJ''»Óée
Xm]tl±á!@ð\ÖzûÚ*Kmä' with RSSI -118 and frequency error of -373
Average RSSI: -118.00

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Figure 1.3: Long Range Test Interference

1.4 GitHub Link

All GitHub files can be found at the following link https://github.com/roryschram/EEE4022S_Final_Project. This includes all code for the project as well as PCB design files.

1.5 Ethics Clearance



PRE-SCREENING QUESTIONNAIRE OUTCOME LETTER

STU-EBE-2023-PSQ000513

2023/08/06

Dear Rory Schram,

Your Ethics pre-screening questionnaire (PSQ) has been evaluated by your departmental ethics representative. Based on the information supplied in your PSQ, it has been determined that you do not need to make a full ethics application for the research project in question.

You may proceed with your research project titled:

LoRa Based Communication System for Radiosonde Applications

Please note that should aspect(s) of your current project change, you should submit a new PSQ in order to determine whether the changed aspects increase the ethical risks of your project. It may be the case that project changes could require a full ethics application and review process.

Regards,

Faculty Research Ethics Committee

Figure 1.4: Ethics Clearance

Bibliography

Glossary